

Programme	BSc Mathematics Honours			
Course Code	MAT2CJ101 / MAT2MN100			
Course Title	INTEGRAL CALCULUS			
Type of Course	Major			
Semester	II			
Academic Level	100-199			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Basic knowledge of Functions, Limits, Continuity and Differentiation (MAT1CJ101 - Differential Calculus).			
Course Summary	The course provides a comprehensive exploration of integral calculus, covering techniques such as indefinite integrals, Riemann sums, definite integrals, properties of integrals, the Fundamental Theorem, L'Hopital's Rule, basic integration formulas, and applications in finding areas between curves, volumes of solids, lengths of plane curves, and areas of surfaces of revolution. Through these topics, students gain proficiency in solving a wide range of mathematical problems involving integration and its applications in various fields.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Solve indefinite and definite integrals of functions.	Ap	F	Internal Exam/Assignment /Seminar/Viva/ End Sem Exam
CO2	Learn logarithmic, exponential, inverse trigonometric functions and to evaluate derivatives and integrals of the above transcendental functions and use it for computations of other limits	U	F	Internal Exam/Assignment /Seminar/Viva/ End Sem Exam
CO3	Apply integration formulas to find the area between two curves, the surface area and volume of a solid of revolution.	Ap	F	Internal Exam/Assignment /Seminar/Viva/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	Calculus and Analytic Geometry, 9th Edition, George B. Thomas, Jr. Ross L. Finney, Pearson Publications, 2010, ISBN: 978-8174906168.			
Module	Unit	Content	Hrs (48+12)	Marks Ext: 70
I	Module I		14	Min.15
	1	Section 4.1 - Indefinite Integrals.		
	2	Section 4.3 - Integration by Substitution - Running the Chain Rule Backward.		
	3	Section 4.5 - Riemann Sums and Definite Integrals. (Example 9 is optional.)		
	4	Section 4.6 - Properties, Area, and the Mean Value Theorem - Topics up to and including Example 6.		
	5	Section 4.6 - Properties, Area, and the Mean Value Theorem- Topics from The Average Value of an Arbitrary Continuous Function onwards.		
II	Module II		11	Min.15
	6	Section 4.7 – The Fundamental Theorem (Example 6 is optional).		
	7	Section 4.8 - Substitution in Definite Integrals.		
	8	Section 6.2 - Natural Logarithms- Topics up to and including The Graph and Range of $\ln x$.		
	9	Section 6.2 - Natural Logarithms. -Topics from Logarithmic Differentiation onwards.		
	10	Section 6.3 - The Exponential Function- Topics up to and including Example 4.		
III	Module III		12	Min.15
	12	Section 6.6 - L' Hopital's Rule		
	13	Section 6.9 - Derivatives of Inverse Trigonometric Functions; Integrals.		
	14	Section 7.1 - Basic Integration Formulas.		
	15	Section 7.2 - Integration by Parts		
IV	Module IV		11	Min.15
	17	Section 5.1 - Areas Between Curves. - Topics up to and including Example 2.		

	18	Section 5.1 - Areas Between Curves- Topics from Boundaries with Changing Formulas		
	19	Section 5.2 - Finding Volumes by Slicing. (Example 2 may be done as open ended).		
	20	Section 5.3 - Volumes of Solids of Revolution- Disks and Washers - Topics up to and including Example 4.		
	21	Section 5.5 - Lengths of Plane Curves. - Topics up to and including Example 2.		
	22	Section 5.6 - Areas of Surfaces of Revolution- Topics up to and including Example 2.		
V	Module V (Open Ended)		12	
	Inverse Functions and their Derivatives, a^x and $\log_a x$, Inverse Trigonometric Functions and their derivatives, Hyperbolic Functions, Integrals and their derivatives, Integration using trigonometric substitutions, Moments and Center of Mass.			

References

1. Howard Anton, Biven, & Stephen Davis, Calculus, 7th Ed., Wiley India

2. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Ed, John Wiley & Sons.

3. Robert T Smith and Roland B Minton, Calculus, 4th Ed. McGraw-Hill Companies

4. Soo T Tan, Calculus, 9th Ed. Brooks/Cole Pub Co.

5. Tom M. Apostol, Calculus, Vol 1: One Variable Calculus with an Introduction to Linear Algebra, 2nd Ed, John Wiley & Sons.

6. Michael Van Biezen Calculus Lectures:
<https://youtu.be/YZYxPclo2rg?si=qKCt6ty8m5dBR4DG>

***Optional topics are exempted for end semester examination**

****70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	1	3	0	3	1	3	0	1
CO 2	2	3	2	1	3	0	3	1	3	0	1
CO 3	2	3	2	1	3	0	3	2	3	0	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓